

Intelligent Agents and AI Environments

Agents, Rationality, PEAS, Agent Types, and Environment Properties

Lecture 2

BS Computer Science — Artificial Intelligence

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ENVIRONMENT

percepts (sensors) →

AGENT

← actions (actuators)

From AI Problems to Intelligent Agents

- AI is ultimately about building systems that act well in the world, not just systems that compute an answer in isolation.
- Nearly every AI system — however different its internals — can be described as something that perceives, decides, and acts.



Real-life example: a self-driving car observes roads, vehicles, pedestrians and traffic signals, then decides and controls the vehicle accordingly.

Systems that fit this pattern

- Autonomous vehicle
- Virtual assistant
- Robot vacuum cleaner
- Recommendation system
- Game-playing AI
- Medical decision-support system

Learning Objectives for Today

- 1 **Define** an intelligent agent and its agent function.
- 2 **Explain** the concept of rationality in an agent.
- 3 **Distinguish** rational agents from merely successful agents.
- 4 **Describe** the PEAS framework for specifying a task.
- 5 **Identify** the five major types of intelligent agents.
- 6 **Classify** task environments using six standard properties.
- 7 **Analyze** real-world AI systems using this vocabulary.
- 8 **Select** an appropriate agent type for a given problem.

What Is an Agent?

Formal definition: an agent perceives its environment through sensors and acts upon that environment through actuators.

ENVIRONMENT



SENSORS



AGENT



ACTUATORS



ENVIRONMENT

Human

- **Sensors** = eyes, ears, skin
- **Actuators** = hands, legs, speech

Autonomous Car

- **Sensors** = cameras, LiDAR, radar, GPS
- **Actuators** = steering, brakes, accelerator

Agent–Environment Interaction



...and the cycle repeats: the changed environment produces a new percept, prompting a new action.

- **Percept** — what the agent senses at one instant.
- **Percept sequence** — the complete history of percepts so far.
- **Action** — what the agent does in response.
- **Environment** — everything external the agent interacts with.

Example: an autonomous vehicle continuously perceives the road, decides, and acts — then perceives the newly-changed road ahead, over and over, many times per second.

Percepts and Percept Sequences

- An agent does not simply receive one isolated input and stop — it receives a stream of percepts over time.

Percept

The agent's current sensory input — a single snapshot.

Percept Sequence

The complete history of every percept the agent has received so far.

Example — a robot vacuum's percept sequence:

Obstacle detected

percept 1



Dirty floor

percept 2



Wall detected

percept 3

Why the past matters: remembering earlier percepts lets an agent infer things the current percept alone cannot reveal — the basis for model-based and learning agents later in this lecture.

Agent Function vs. Agent Program

$$f : \text{Percept Sequences} \rightarrow \text{Actions}$$

- **Agent function** — the abstract mapping from percept sequences to actions (what SHOULD be done).
- **Agent program** — the actual code or circuit that computes that mapping (HOW it is done).

Percept Sequence



Agent Function



Action

Example: a thermostat's agent FUNCTION says "below 20°C → heat ON"; its agent PROGRAM is the specific circuit or code that implements that rule.

What Makes an Agent “Intelligent”?

Characteristics often associated with intelligence

- Perception
- Decision making
- Goal-directed behavior
- Adaptation
- Learning
- Autonomy
- Responsiveness

Important: an agent need NOT possess every trait above. What counts as "intelligent" depends on the task and the agent's design.

Example: a thermostat has very limited intelligence compared with an autonomous vehicle, yet both are legitimately agents — they simply sit at different points on a spectrum of capability.

What Is a Rational Agent?

Formal definition: a rational agent chooses the action expected to maximize its performance measure, given the percept sequence and available knowledge.

Rational $\neq$ **always successful**

Rational $\neq$ **omniscient**

Rational $\neq$ **perfect**

Real-life example: a self-driving car chooses to brake because current sensor information indicates a pedestrian ahead — that is the rational choice given what it currently knows.

Rationality is about making the best decision with the information available — not about guaranteeing a good outcome.

Rationality Depends on Four Factors

$$\text{Rational Action} = f(\text{Performance Measure}, \text{Prior Knowledge}, \text{Available Actions}, \text{Percept History})$$

1. Performance measure

What counts as success? — e.g. safety and travel time.

2. Prior knowledge

What the agent already knows about the world.

3. Available actions

What the agent is physically able to do.

4. Percept sequence

Everything the agent has sensed up to now.

Self-driving car: safety+comfort (measure), knows traffic rules (knowledge), can steer/brake/accelerate (actions), just saw a pedestrian (percepts) — together these determine the rational action.

Rationality vs. Omniscience

Rational Agent

Makes the best decision based on available information.

- Judged at the moment of the decision
- Can still turn out unlucky

Omniscient Agent

Would already know the actual future outcome.

- An idealization — impossible in reality
- Never wrong, by definition

Example: a self-driving car slows down because a hidden obstacle MAY be ahead. If the obstacle turns out not to exist, the decision to slow down can still have been rational.

Rationality is based on available information, not hindsight.

Performance Measure

- Agents need an explicit criterion for evaluating success — without one, "rational" has no meaning.

Autonomous Vehicle

- Safety
- Travel time
- Comfort
- Energy efficiency
- Traffic-law compliance

Robot Vacuum

- Cleanliness
- Time
- Energy use
- Coverage

Recommendation System

- User satisfaction
- Relevance
- Engagement
- Long-term usefulness

Key point: a good agent design **REQUIRES** a well-defined performance measure — vague goals produce vague (or dangerously wrong) behavior.

Worked Example — A Delivery Robot's Dilemma

Situation — the robot currently detects:

- Pedestrian nearby
- Low battery
- A road crossing ahead
- An approaching delivery deadline

Classroom question: “What should the robot do right now?”

- **Performance measure** — how are safety, punctuality, and battery use weighed against each other?
- **Percepts** — pedestrian, crossing, battery level, and deadline, all sensed right now.
- **Available actions** — wait, proceed slowly, reroute, or return to base to recharge.
- **Prior knowledge** — traffic rules, the map, and typical pedestrian behavior.

Introduction to PEAS

- PEAS is a systematic way to specify any agent's task environment precisely — so designs can be described, compared, and evaluated consistently.

P Performance Measure

How do we judge whether the agent is succeeding?

E Environment

What surrounds the agent, and is affected by or affects it?

A Actuators

How does the agent act upon its environment?

S Sensors

How does the agent perceive its environment?

P — Performance Measure

Definition

The performance measure is the criterion used to judge how successful the agent's behavior is — it defines what “success” means for this task.

Autonomous Taxi

- Safety
- Passenger satisfaction
- Travel time
- Fuel / energy efficiency
- Legal compliance

Robot Vacuum

- Clean floor
- Coverage
- Time taken
- Battery efficiency

Key point: the performance measure is what defines success for an agent.

E — Environment

Definition

The environment is everything external to the agent that it interacts with — what counts as “the environment” is entirely task-dependent.

Autonomous Car

- Roads and lanes
- Other vehicles
- Pedestrians
- Traffic lights & signs
- Weather conditions

Chess-Playing Agent

- The chessboard
- The pieces
- The opponent
- The rules of chess

Key point: what counts as “the environment” is entirely task-dependent.

A — Actuators

Definition

Actuators are the mechanisms through which an agent acts upon, and produces changes in, its environment.

General-Purpose Robot

- Wheels or legs
- Robotic arms
- Grippers

Autonomous Car

- Steering
- Brakes
- Accelerator

Virtual Assistant

- Speech output
- Screen display
- Notifications

S — Sensors

Definition

Sensors are the mechanisms through which an agent perceives its environment and receives percepts.

Autonomous Car

- Cameras
- Radar
- LiDAR
- GPS

Robot Vacuum

- Bump sensors
- Distance sensors
- Cameras

Virtual Assistant

- Microphone
- Text input
- Camera (where available)

Complete PEAS Example: Autonomous Taxi

PEAS Element	Description
P — Performance Measure	Safety, passenger comfort, travel time, energy efficiency, legal compliance
E — Environment	Roads, other traffic, pedestrians, weather, passengers
A — Actuators	Steering, accelerator, brake, horn, indicators
S — Sensors	Cameras, radar, LiDAR, GPS, speed sensors

A well-formed PEAS description should let a new engineer understand the whole task without seeing any code.

PEAS Examples from Real Life

Robot Vacuum

P: Clean floor, coverage, time, battery use

E: Home interior, furniture, obstacles

A: Wheels, brushes, suction motor

S: Bump, cliff, and dirt sensors

Medical Diagnosis Assistant

P: Diagnostic accuracy, patient safety

E: Patient records, symptoms, test results

A: Recommendations, alerts, reports

S: Data entry, lab results, imaging

Chess-Playing Agent

P: Win rate, move quality

E: Chessboard, pieces, opponent, rules

A: Piece moves

S: Board-state input

Online Recommendation Agent

P: User satisfaction, relevance, engagement

E: User history, catalog, browsing behavior

A: Displayed recommendations

S: Clicks, ratings, browsing data

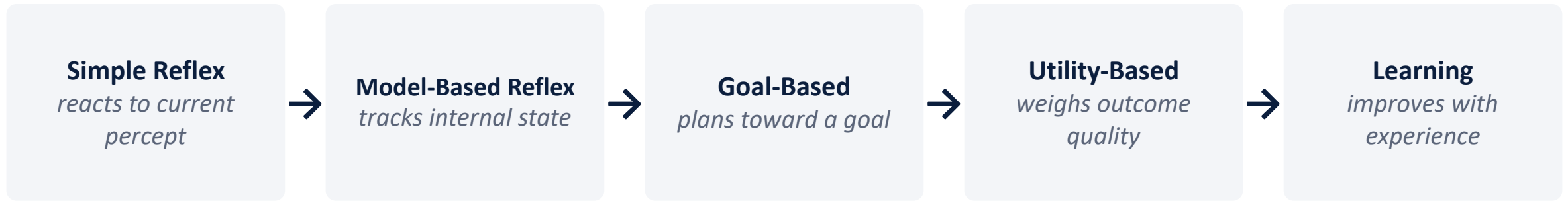
Classroom Activity: Design PEAS

Scenario: Food Delivery Drone

- **P — Performance:** what defines success for this drone?
- **E — Environment:** what does it operate in?
- **A — Actuators:** how does it act?
- **S — Sensors:** how does it perceive its surroundings?

Discuss in pairs for 2-3 minutes, then share your PEAS description with the class.

Overview of Agent Types



Important: increasing sophistication does NOT mean “always better.” The right agent type depends on the task, not on how advanced it sounds.

Example: a basic thermostat (simple reflex) does its one job perfectly well — building a learning agent to control it would be needless complexity and cost.

Simple Reflex Agents

Definition

A simple reflex agent selects its action based only on the CURRENT percept — it has no memory of anything sensed before.

Current Percept



Condition



Action

Rule: IF temperature < threshold THEN heater = ON

Real-life examples: basic thermostats, automatic doors, motion-triggered lights, basic obstacle avoidance.

Strength: fast, cheap, and simple to build. **Limitation:** no memory — struggles when the environment is only partially observable.

Simple Reflex Agent: A Detailed Example

Scenario: an automatic door



Rule: IF person detected THEN open door, ELSE close door

- **Why no world model is needed** — the correct action depends only on whether a person is detected RIGHT NOW, never on what was detected earlier.
- **Limitation to discuss** — what should the door do if the motion sensor briefly fails or misfires?

Model-Based Reflex Agents

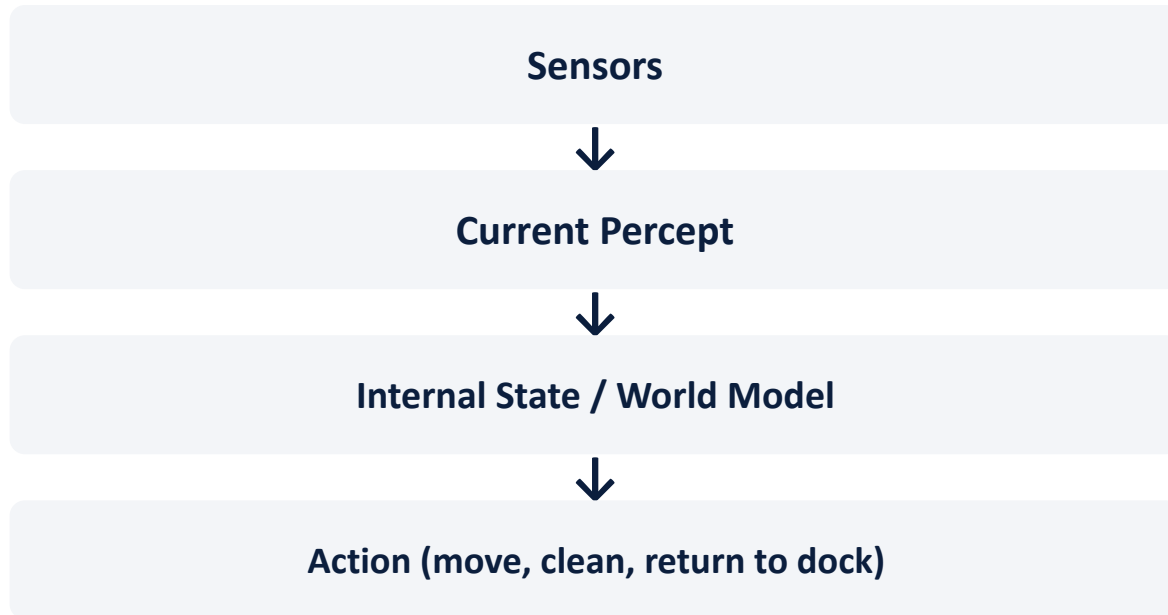
- Simple reflex agents fail when the environment is only PARTIALLY OBSERVABLE — the current percept alone does not tell the whole story.



Example: a robot vacuum that remembers

- Which areas it has already cleaned (avoids repeating them)
- Where obstacles and furniture are located
- Its approximate map of the room, built up over time

Model-Based Agent: A Real-Life Example



Simple Reflex Robot

- Reacts to bump/obstacle only
- May revisit the same spot repeatedly
- No memory of the room

Model-Based Robot

- Remembers visited areas
- Builds an approximate map
- Tracks battery state over time

Goal-Based Agents

Definition

A goal-based agent considers the FUTURE consequences of its actions, choosing ones that are expected to achieve a desired goal.



Example — GPS navigation: the goal is “reach the destination.” The system considers multiple alternative routes and selects one expected to achieve that goal.

Key contrast: a model-based agent asks “what is true now?”; a goal-based agent also asks “what sequence of actions gets me where I want to be?” — far more flexible than fixed rules.

Goal-Based Agent: Real-Life Examples

System	Current State	Goal	Possible Actions
GPS route planning	Current location	Reach destination	Alternative routes
Chess-playing AI	Board position	Checkmate opponent	Legal moves
Warehouse robot	Robot's position	Deliver item to bay	Move, turn, lift
Puzzle-solving system	Current arrangement	Solved configuration	Valid moves

GPS, fully worked: starting from the current location, the system searches over alternative routes and selects the one expected to reach the destination — this planning process connects directly to the search algorithms in the next lecture.

Utility-Based Agents

- **Why goals alone are not enough** — two different actions can both achieve the same goal. Which one is actually BETTER?

Utility = a numeric measure of how DESIRABLE an outcome is, not merely whether it was achieved.



Example — GPS again: several routes all reach the destination (the goal). A utility-based GPS also weighs speed, cost, and safety to choose the BEST one, not just any working one.

Goal = did we get there? Utility = how good was the way we got there?

Utility-Based Agent: A Real-Life Example

Route A

- Faster travel time
- More traffic congestion
- Higher accident risk

Route B

- Slower travel time
- Safer road conditions
- Lower congestion

Both routes reach the destination — the goal is satisfied either way.

- **Goal** (binary): reach the destination — both routes succeed equally.
- **Utility** (a matter of degree): weighs speed against safety and congestion — the two routes are NOT equally desirable.

Learning Agents

- A learning agent improves its behavior over time from experience, rather than relying only on fixed, hand-designed rules.

Performance Element

Selects actions — this is essentially the “agent” from the earlier architectures.

Learning Element

Makes improvements based on feedback about how well the agent is doing.

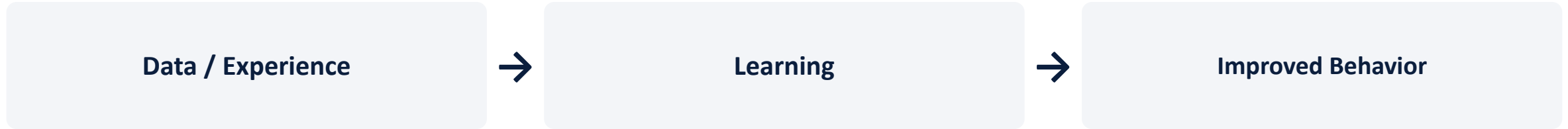
Critic

Evaluates the agent's behavior against a fixed performance standard and gives feedback.

Problem Generator

Suggests exploratory actions that lead to new, potentially useful experiences.

Learning Agent: Real-Life Examples



- Recommendation systems — improve suggestions from viewing/purchase history
- Spam filters — improve accuracy as users mark more mail as spam or not
- Autonomous driving systems — refine perception and driving policy from data
- Voice assistants — adapt to a speaker's accent and vocabulary over time
- Game-playing agents — improve strategy through repeated play

We are not covering the specific learning algorithms behind these systems today — only the architectural pattern.

Compare the Five Agent Types

Agent Type	Main Idea	Memory?	Goals?	Utility?	Learns?
Simple Reflex	Condition → action rules	No	No	No	No
Model-Based Reflex	Rules + internal state	Yes	No	No	No
Goal-Based	Plans toward a goal	Yes	Yes	No	No
Utility-Based	Weighs outcome desirability	Yes	Yes	Yes	No
Learning	Improves from experience	Yes	Usually	Usually	Yes

No row is “the best” in general — each is the right tool for a different kind of task.

What Is a Task Environment?

- A task environment is defined along six independent dimensions — together they describe the nature of the problem an agent faces.

Fully Observable
↔
Partially Observable

Deterministic
↔
Stochastic

Episodic
↔
Sequential

Static
↔
Dynamic

Discrete
↔
Continuous

Single-Agent
↔
Multi-Agent

Environments are rarely purely one side — most real systems combine several properties simultaneously (see slide 40).

Fully Observable vs. Partially Observable

Fully Observable

The agent's sensors give access to the complete state relevant to its decision.

- A chess board in plain view

Partially Observable

Some relevant information is hidden, noisy, or simply unavailable to the sensors.

- Driving in fog, medical diagnosis, poker, a robot navigating behind obstacles

Important: very few real sensors are perfect — always ask “fully or partially observable?” rather than assuming perfect sensing.

Why is a self-driving car's environment partially observable?

Deterministic vs. Stochastic

Deterministic

The next state is completely determined by the current state and the agent's action.

- Sudoku, chess (the rules themselves)

Stochastic

Outcomes involve genuine uncertainty or probability — the same action can lead to different results.

- Real traffic, weather prediction, medical treatment outcomes, sensor/actuator noise

Key point: most real-world environments are stochastic, at least to some degree — even carefully engineered robots face sensor and actuator uncertainty.

Classroom question: Is chess deterministic?

Episodic vs. Sequential

Episodic

Each decision is a self-contained episode, largely independent of previous decisions.

- Independent image classification, independent product-quality inspection

Episode 1
independent



Episode 2
independent



Episode 3
independent

Episodic: no arrows carry meaning between episodes.

Sequential

Current decisions affect future states and future decisions — order and history matter.

- Chess, driving, robot navigation, holding a conversation

Move 1



Move 2



Move 3

Sequential: each move constrains what comes next.

Static vs. Dynamic

Static

The environment does not change while the agent is deciding what to do.

- A crossword puzzle, an offline board configuration

Dynamic

The environment can change WHILE the agent is still deciding.

- Autonomous driving, air traffic control, robot navigation, online trading

A useful middle case — “semidynamic”: the environment itself does not change, but the agent's PERFORMANCE SCORE does (e.g. a chess clock ticking while a player thinks).

Dynamic environments punish slow decision-making — delay itself has a cost.

Discrete/Continuous & Single/Multi-Agent

PANEL A — DISCRETE VS. CONTINUOUS

Discrete

- A finite number of distinct states and actions.
- Chess, tic-tac-toe

Continuous

- States and actions vary smoothly over a continuous range.
- Robot-arm movement, autonomous driving, aircraft control, temperature control

PANEL B — SINGLE-AGENT VS. MULTI-AGENT

Single-Agent

- Only one agent's actions affect its own performance.
- Sudoku solver, maze-solving robot

Multi-Agent

- Multiple agents interact — cooperating, competing, or both at once.
- Chess, football-playing robots, vehicles sharing roads, online auctions

Putting Everything Together

Agent

The autonomous driving agent — continuously perceives, decides, and acts.

PEAS

P: safety, comfort, efficiency
E: roads, traffic, weather
A: steering, brakes, accel.
S: cameras, radar, LiDAR, GPS

Agent Type

A combination of model-based, goal-based, utility-based, and learning components together.

Environment

Partially observable, stochastic, sequential, dynamic, continuous, and multi-agent — all at once.

Environment



Sensors / Percepts



Agent (Decision)



Actuators



Environment

The loop repeats continuously as the vehicle drives — this is the conceptual map from Section B, now fully instantiated.

Final Classroom Challenge:

Choose ONE real-world AI system and identify (1) its PEAS description, (2) its agent type, and (3) the properties of its environment.

Examples: smart home assistant, robot vacuum, medical diagnosis system, autonomous vehicle, recommendation system, delivery drone.

Next: *how can an agent systematically find a sequence of actions to achieve a goal? → Search and Problem Solving.*